

Analyzing Long-Horizon Failures of LLM Agents on OS Tasks

Team 3: Skylar Hou, Linxiang Jin, Xinyu Jessica Wang, Kris Yang, Handan Hu

Brief Recap

- **Problem**

As task horizon (complexity) grows, agent performance drops sharply
Understanding where and why long-horizon OS agents break is important

- **Datasets**

AgentBench OS Interaction Tasks
Evaluated by - Task Success Rate

Baseline Methods

- **AgentBench: Evaluating LLMs as Agents.**
 - Introduces an interactive benchmark with 8 environments (OS, DB, web tasks...) requiring multi-step actions.
 - Results show strong models still struggle as task horizon increases.
- **The Long-Horizon Task Mirage.**
 - Proposes the HORIZON framework
 - Defining intrinsic horizon and a failure taxonomy (planning errors, forgetting, etc.) to diagnose why agent systems break on long tasks.
- **Our Goal**
 - Replicate long-horizon OS agent results from AgentBench
 - Extend existing OS tasks with longer intrinsic horizon and evaluate
 - Categorize failure types based on new OS tasks

Experiment Results - Replication

Experiment setup

- **Hardware:** Google Cloud n2-standard-4 (4 vCPUs, 16 GB Memory)
MacOS M3 (16 vCPUs, 24 GB Memory)
- **Models:** API-based Commercial LLMs (e.g. GPT-5-mini)
- **Framework:** Reuse **AgentBench** for evaluation, 144 OS tasks

Findings

GPT-5-mini:

- **Average success rate** (4 runs): **22.7% $\pm$ 2.1%**
- **Best-of-n:** up to **34.0% (pass@4)**
- **15 / 144** tasks are consistently solved, which can be used as baseline tasks for controlled horizon extension

Findings - Replicate Results from AgentBench

Temperature = 0 Runs = 2	Our Experiment (Average succ rate in %)	Result from AgentBench Paper (%)	Difference (%)
gpt-4	37.2	42.4	-5.2
claude-3	16.0	22.9	-6.9
gpt-3.5-turbo	23.6	32.6	-9.0
claude-sonnet-3.7	48.3	36.5	+11.8
gpt-4o	36.8	38.5	-1.7
o3-mini	35.4	35.1	+0.3
o4-mini	45.1	41.8	+3.3

Findings - Failure Classification

- Use DeepSeek **LLM as a judge** to classify each failed execution trace into 1 of 7 predefined failure types
- Construct a structured prompt that includes the failure taxonomy, label definitions, and decision criteria for consistent classification
- Insert each failed execution trace into the prompt and ask the model to return exactly one failure label

task_index	classification_type
0	Environment Disturbance / Unable to Detect Change
1	Instruction Error
2	Instruction Error
3	Environment Disturbance / Unable to Detect Change
4	Environment Disturbance / Unable to Detect Change
5	Environment Disturbance / Unable to Detect Change
6	False Assumptions
7	Environment Disturbance / Unable to Detect Change
8	Environment Disturbance / Unable to Detect Change
9	Environment Disturbance / Unable to Detect Change
10	Environment Disturbance / Unable to Detect Change

Findings - Failure Classification

Prompt Input: 1 task + 1 failed trace

You are an expert judge for failure analysis of LLM agents on long-horizon tasks.

You will receive: - the original task description - one failed execution trace from the agent Your job is to classify the primary root cause of failure into exactly one label from this closed set:

1. Environment Disturbance / Unable to Detect Change Definition: The environment changes or rejects an action, but the agent fails to notice or properly react. Example: The agent misses a permission denial and continues issuing commands based on a false belief of success.
2. Instruction Error Definition: The agent misunderstands or ignores explicit task instructions, constraints, or exception clauses. Example: The agent applies a recursive command globally, ignoring specific exception clauses in the prompt.
3. Catastrophic Forgetting Definition: The agent forgets an important earlier constraint, goal, or fact and later violates it in a serious way. Example: The agent forgets an earlier constraint to not modify sensitive files.
4. False Assumptions Definition: The agent acts on an unverified belief about the environment, files, users, permissions, or task state. Example: The agent assumes ownership by the system user without actually verifying ownership.
5. Planning Errors (Sub-plan & Action) Definition: The agent forms an incorrect sub-plan or chooses an inappropriate action sequence even if it understood the instruction. Example: Applying a blanket permission change instead of first identifying the eligible target files.
6. History Error Accumulation Definition: An earlier mistake propagates because the agent treats a failed or invalid earlier step as if it succeeded, causing downstream errors. Example: An invalid command fails silently, but the agent assumes success and issues dependent commands, cascading the error.
7. Memory Limitations Definition: The agent loses access to earlier relevant context because of context-window or recall limitations.

7 taxonomies

Example: Important earlier notices about file ownership are truncated from the context window.

Rules:

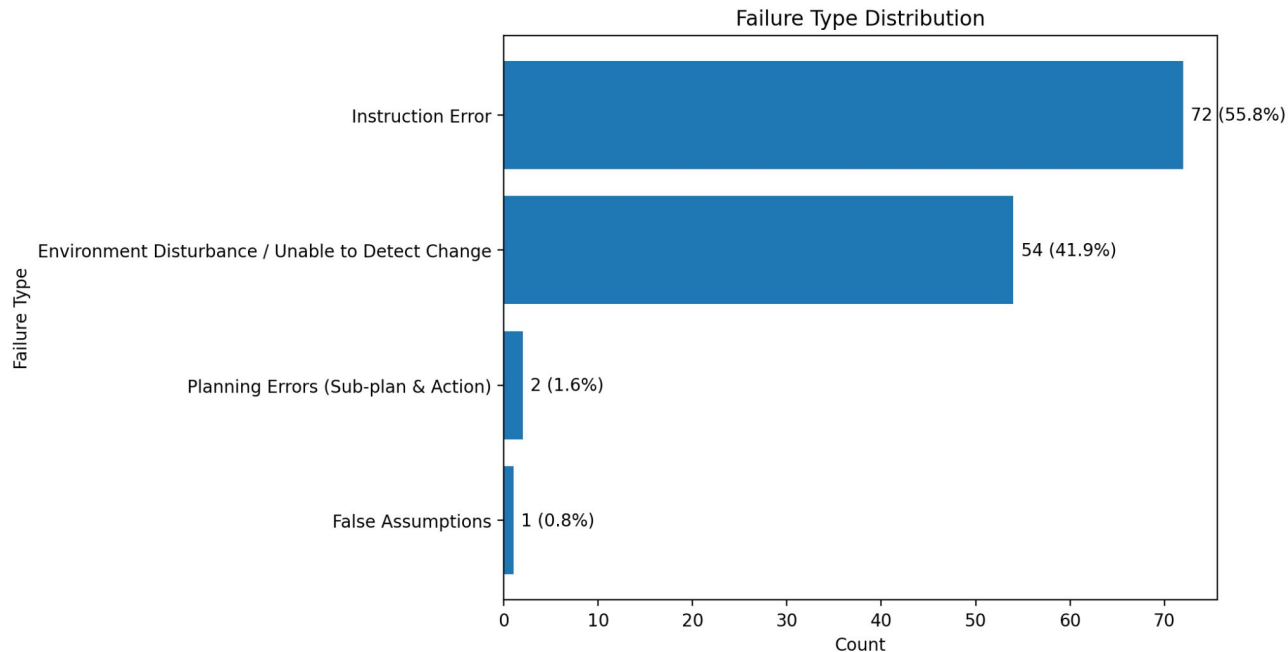
- Return exactly one label from the closed set above.
- Choose the single best primary root cause, not multiple labels.
- Prefer the earliest root cause that best explains the downstream failure.
- Use both the task description and the trace.
- If the agent simply misunderstood the task requirements, choose Instruction Error.
- If the agent understood the task but picked a bad method or ordering, choose Planning Errors (Sub-plan & Action).
- If the agent explicitly relies on something it never checked, choose False Assumptions.
- If the agent forgets an earlier instruction or fact that was previously available in the trace, choose Catastrophic Forgetting.
- If the trace suggests the needed earlier information dropped out of context or was no longer accessible, choose Memory Limitations.
- If an earlier failed step causes later dependent mistakes because the agent assumes success, choose History Error Accumulation.
- If the environment gives feedback like permission denied, missing file, changed state, or other disturbance and the agent fails to detect/adapt, choose Environment Disturbance / Unable to Detect Change.

Rules for classification

Output format: Return only the label text, with no explanation, no JSON, and no extra words.

Output: 1 label

Findings - Failure Analysis on GPT5-mini Model



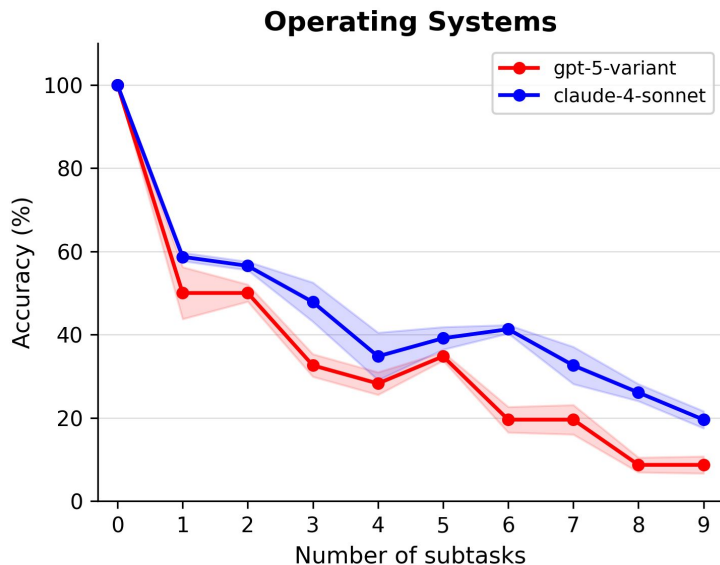
The remaining categories, including Catastrophic Forgetting, History Error Accumulation, and Memory Limitations, do not appear in the result.

Possible reason: short-horizon tasks, incomplete trace information

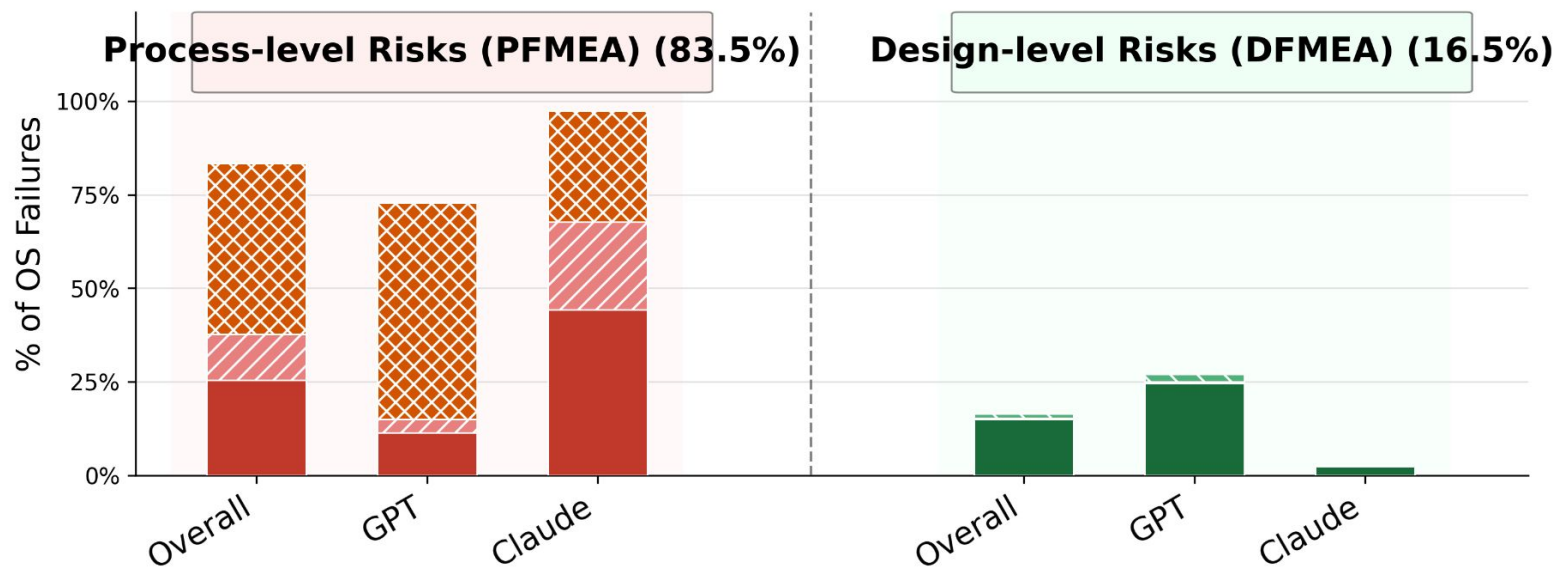
Experiment Results - Extention

Experiment setup

- **Tasks:** Selected **34** tasks from agentbench 100% average success rate
- **Models:** GPT-5-mini & Claude-4-Sonnet



Findings - Failure Classification



Failure Modes:

- | | | | |
|-------------|----------------------------|-------------------------|------------------|
| Environment | Planning Error | Memory Limitation | False Assumption |
| Instruction | History Error Accumulation | Catastrophic Forgetting | |

Future Plan

Future Extension

- Construct long-horizon OS tasks by extending verified baseline tasks
- Analyze performance trends with increasing task horizon
- Refine failure classification to improve annotation accuracy

Contribution Statement

- Handan Hu – Experiment setup (20%)
- Skylar Hou – Model evaluation (20%)
- Xinyu Jessica Wang – Task construction (20%)
- Kris Yang – Failure classification (20%)
- Linxiang Jin – Result analysis (20%)